Numerical Simulation of Heat Distribution and Energy Efficiency in Residential Buildings

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Abstract

This study investigates the dynamics of heat transfer within a multi-room apartment using a numerical solver for the 2D Heat Equation. By implementing a Finite Difference Method (FDM), we analyze the impact of radiator placement and the effectiveness of time-based heating schedules across various external temperature profiles. Our results indicate that placing radiators under windows enhances thermal uniformity and that strategic heating interruptions can lead to energy savings of up to 29% in moderate weather conditions.

1 Introduction

Effective thermal management is a key factor in sustainable building design. This project aims to simulate heat flow using partial differential equations (PDEs) to evaluate how different heating strategies influence energy consumption and occupant comfort. The simulation accounts for external temperature fluctuations, building geometry, and the physical properties of air.

2 Mathematical Framework

2.1 Governing Equation

The temperature evolution $u(x, t)$ is modeled using the 2D Heat Equation with a source term representing the radiators:

$$\frac{\partial u}{\partial t} = \alpha \Delta u + \frac{P}{\rho A c} \Theta(x, u) \tag{1}$$

Where:

- α is the thermal diffusivity coefficient.
- P is the heating power (800 W).
- ρ is the air density ($1.2\,\mathrm{kg/m^3}$).
- A is the area and c is the specific heat capacity ($1005\,\mathrm{J/(kg \cdot K)}$).
- $\Theta(x, u)$ is an indicator function for radiator activity, governed by a thermostat logic.

2.2 Boundary Conditions

To close the system, we define the following conditions:

1. **Dirichlet Condition (Windows):** $u(x, t) = T_{out}(t)$, where T_{out} is the time-dependent external temperature.
2. **Neumann Condition (Walls):** $\nabla u \cdot \mathbf{n} = 0$, assuming ideal insulation with zero heat flux.

3. **Initial Condition:** $u(x, 0) = u_0(x)$ representing the starting temperature distribution (18°C).

3 Numerical Implementation

The simulation employs the Finite Difference Method (FDM) on a discretized grid. The Laplacian operator is approximated using the standard five-point stencil:

$$\Delta u \approx \frac{u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{i,j}}{h_x^2} \tag{2}$$

For nodes located on the boundaries ($\partial\Omega$), the stencil is modified to reflect the insulation properties (reflecting boundary conditions) or the fixed external temperature. The time step is set to $h_t = 3$ minutes, ensuring stability and capturing the daily temperature cycle over a 24-hour period.

4 Assumptions and Parameters

- **Thermostat Control:** Heaters automatically deactivate when the room center reaches the target temperature (21°C).
- **Inter-room Dynamics:** Doors are modeled as open passages where temperatures are averaged at the interface to simulate natural convection.
- **External Data:** Temperature profiles are categorized into three scenarios: *Cool* (10-15°C), *Cold* (2-7°C), and *Very Cold* (-5 to 0°C).

5 Case Studies and Results

5.1 Optimal Radiator Placement

We compared two configurations: radiators placed under windows versus on internal walls.

- **Sub-window Placement:** Acts as a thermal barrier against cold air infiltration, leading to a more homogeneous temperature distribution and more frequent thermostat cycling (energy efficiency).
- **Internal Wall Placement:** Results in significant temperature gradients, with cold zones persisting near the windows, reducing overall comfort.

5.2 Energy Efficiency of Heating Schedules

We tested the economic viability of turning off the heating for 6 hours (10:00–16:00) daily. The cumulative energy usage compared to constant heating is summarized in Table 1.

Table 1: Energy consumption and relative savings with a 6-hour daily heating interruption.

External Condition	Relative Energy Used (%)	Effective Savings (%)
Cool (10°C to 15°C)	71%	29%
Cold (2°C to 7°C)	80%	20%
Very Cold (-5°C to 0°C)	92%	8%

The data suggests that while "turning off the heat" is always beneficial, the relative savings diminish significantly in extreme cold. This occurs because the thermal inertia of the apartment requires more energy to recover the target comfort temperature after the interruption.

6 Conclusions

The numerical model confirms that radiator placement significantly impacts thermal comfort. Furthermore, implementing a heating schedule is an effective strategy for reducing energy consumption, particularly in moderate climates. Future iterations of this model could incorporate 3D heat flow and humidity factors for higher fidelity.